Project Size.	5.0 inch
Model No.	P050H031-V2-IPS-CTP
Samples No.	
Product type.	720xRGBx1280 MIPI mode
Signature by customer:	

Prepared	Checked	Approved

Shenzhen Polcd Technology Co.,Limited

Email: polcd@polcd.com

Mobile: 86-136 0019 7172

1.0 GENERAL DESCRIPTION

Item	Specification	Unit
Screen Size	5.0 inch	Diagonal
Number of Pixel	720RGB(H)x1280(V)	Pixels
Display area	62.10(H)x110.40(V)	mm
Pixel pitch	0.0945(H)x0.0945(V)	mm
Outline Dimension	78.56x135.65x3.1	mm
Pixel arrangement	RGB Vertical Stripe	— —
Display mode	Normally Black	— —
Viewing Direction(eye)	ALL	— —
Gray inversion direction	--	
Display Color	262K	— —
Luminance(cd/m^2)	350	nit
Contrast Ratio	800:1	— —
Surface treatment	— —	— —
Interface	MIPI	
Back-light	LED Side-light type	— —
LCM IC	ST7703	
CTP IC	GT911	
Operation Temperature	-20~70	℃
Storage Temperature	-30~80	℃
Weight	— —	g

1.1 Features

- MIPI interface.

1.2 Applications

- MPOS Device.
- Personal Navigation Device.
- Other devices which require high quality displays.

2.0 INPUT INTERFACE PIN ASSIGNMENT

FPC connector is used for electronics interface

PinNo.	Symbol	Function
1	NC	NC
2	LEDK	LED back light(Cathode)
3	NC	NC
4	LEDA	LED back light(Anode)
5	NC	NC
6	GND	Ground
7	MIPI_D0N	MIPI DSI differential data
8	MIPI_D0P	MIPI DSI differential data
9	GND	Ground
10	MIPI_D1N	MIPI DSI differential data
11	MIPI_D1P	MIPI DSI differential data
12	GND	Ground
13	MIPI_CLN	MIPI DSI differential clock
14	MIPI_CLP	MIPI DSI differential clock
15	GND	Ground
16	MIPI_D2N	MIPI DSI differential data
17	MIPI_D2P	MIPI DSI differential data
18	GND	Ground
19	MIPI_D3N	MIPI DSI differential data
20	MIPI_D3P	MIPI DSI differential data
21-22	GND	Ground
23	NC	NC
24	GND	Ground
25	TE	Frame synchronization signal
26	RESET	external reset input
27	IOVCC(1.8V)	Power Supply 1.8V
28	VDD(2.8V)	Power Supply 2.8V
29-30	GND	Ground

CTP connector is used for electronics interface

1	RST	Touch screen reset
2	SDA	Touch screen data signal
3	SCL	Touch screen clock signal
4	INT	Touch screen interrupt signal
5	VDD(2.8V)	Power Supply 2.8V
6	GND	Ground

3.0 OPTICAL CHARACTERISTICS

3.1 Optical specification

Item		Symbol	Condition	Min	Type	Max	Unit	Note
White luminance (Center)		Lv	Θ=0 Normal Viewing Angle I_{BL}=40mA	--	350	--	cd/m^2	(4)(5)(7)
Response time		Tr+Tf		--	10	15	ms	(1)(3)
Contrast ratio		CR		640	800	--	--	(1)(2)
Color Chromaticity (CIE1931)	white	Wx		0.283	0.303	0.323		(1)(4)
		Wy		0.303	0.323	0.343		
Viewing Angle	Hor	ΘL	CR≥10	--	80	--		(1) (4) Measuring with Polarizer, Reference Only
		ΘR		--	80	--		
	Ver	ΘU		--	80	--		
		ΘD		--	80	--		
Brightness uniformity		Avg	Θ=0	80	90	--	%	(5)
Color Gamut		NTSC	Θ=0	--	70	--	%	C-light
Optima View Direction		ALL						(5)

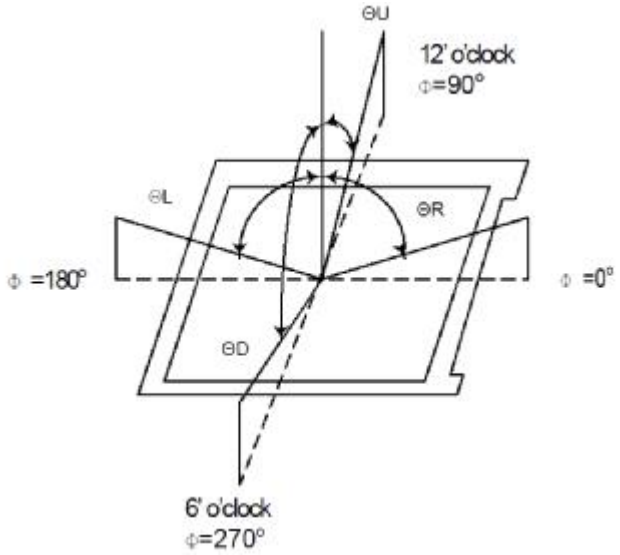
3.2 Measuring Condition

- Measuring surrounding: dark room
- LED current IL:40mA
- Ambient temperature: 25±2℃
- 15min. warm-up time

3.3 Measuring Equipment

- FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-7 for other optical characteristics.
- Measuring spot size：20 ~ 21 mm

Note (1) Definition of Viewing Angle

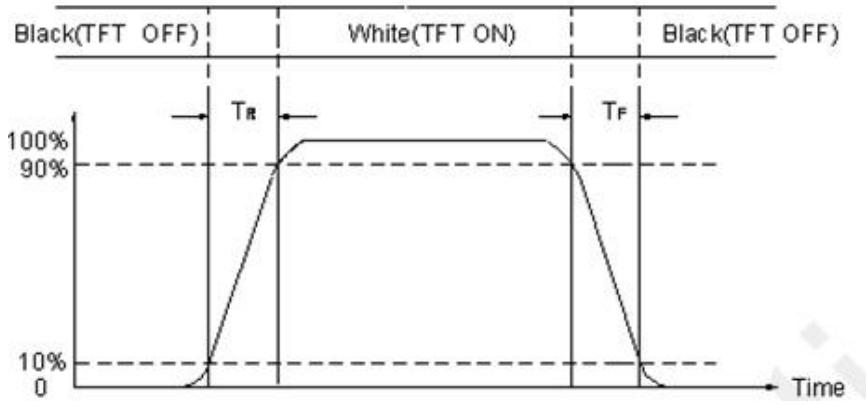


Note (2) Definition of Contrast Ratio(CR):

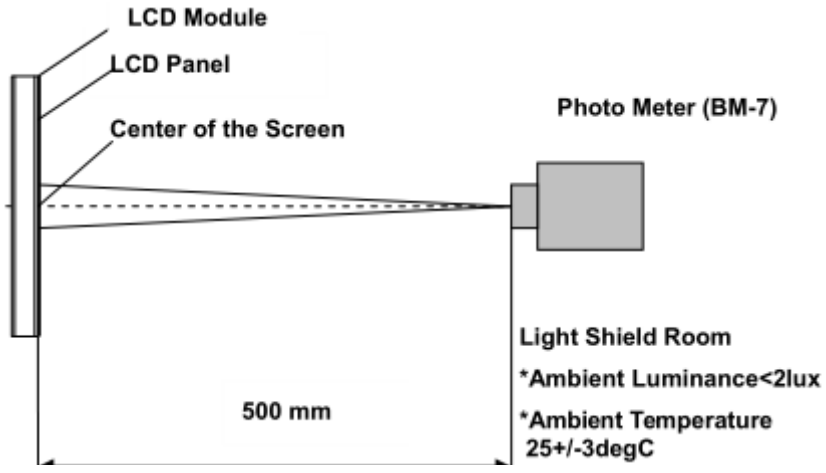
Measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

Note (3) Definition of Response Time: Sum of TR and TF



Note (4) Definition of optical measurement setup



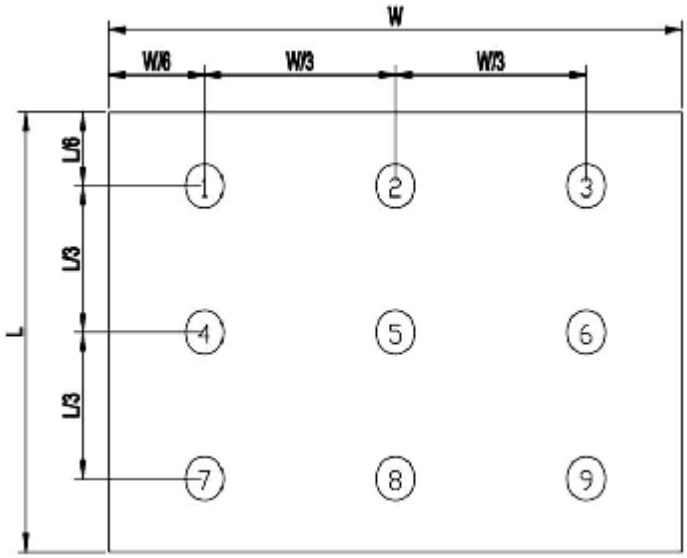
Note (5) Definition of brightness uniformity

The luminance uniformity is calculated by using following formula.

△Bp = Bp (Min.) / Bp (Max.)×100 (%)

Bp (Max.) = Maximum brightness in 9 measured spots

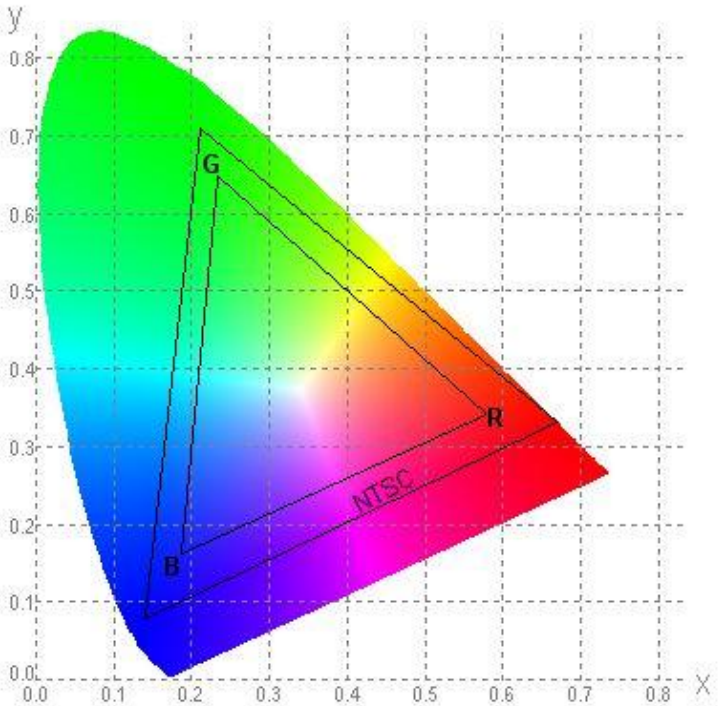
Bp (Min.) = Minimum brightness in 9 measured spots .



Note (6) Definition of Color of CIE1931 Coordinate and NTSC Ratio.

Color gamut:

$$S = \frac{\text{Area of RGB triangle}}{\text{Area of NTSC triangle}} \times 100\%$$



Note (7) Measured the luminance of white state at center point.

4.0 ELECTRICAL CHARACTERISTICS

4.1 TFT LCD Module

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
Analog supply voltage	VDD	2.4	2.8	3.3	V	
Digital supply voltage	VDDI	1.65	1.8	3.3		
Input signal Voltage	VIH	0.7VDDI	-	VDDI	V	
	VIL	GND	-	0.3VDDI	V	

6.2 Back-Light Unit

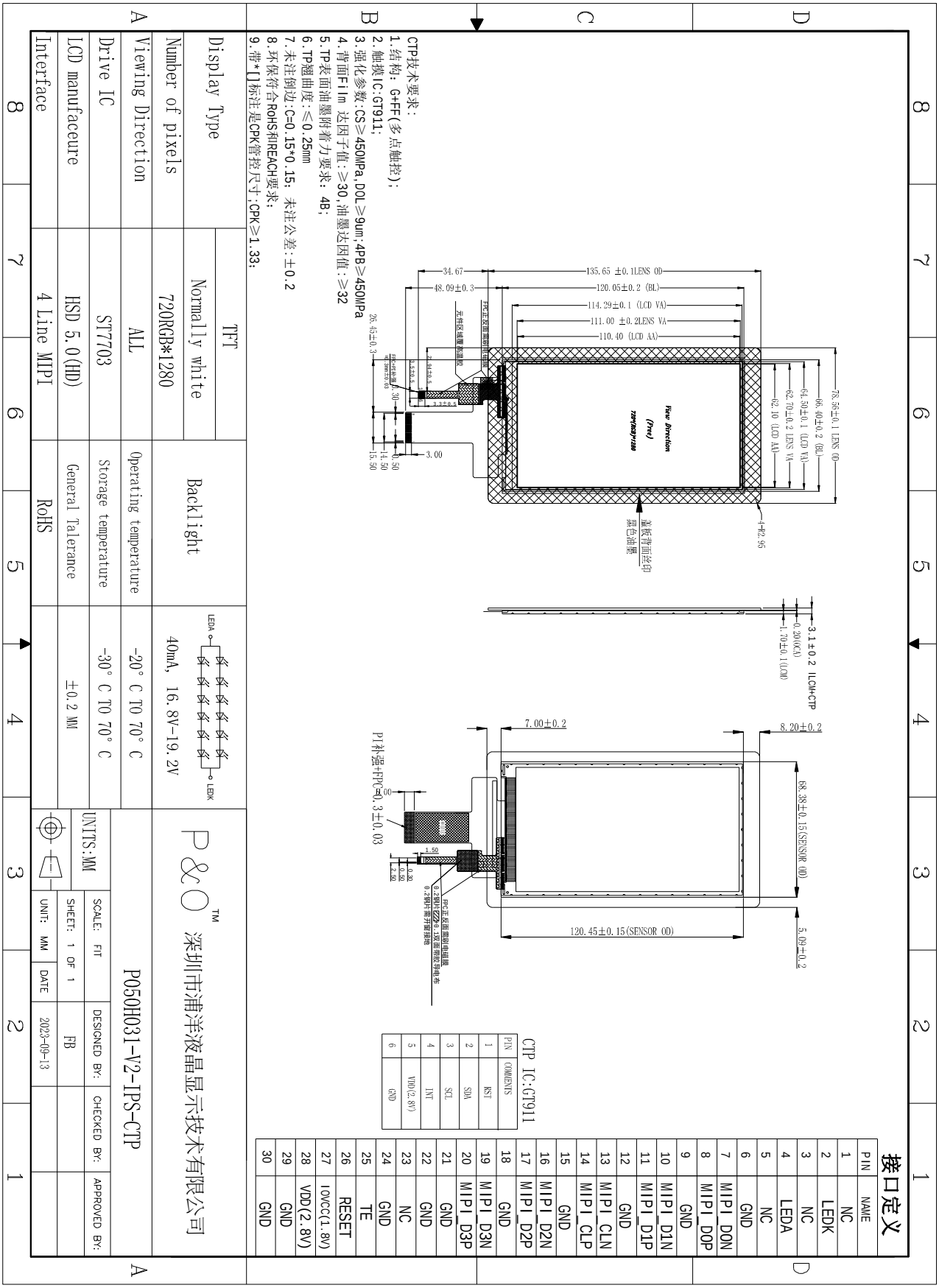
The backlight system is an edge-lighting type with 12 LED Dies.
The characteristics of the LED are shown in the following tables.

Item	Symbol	Min	Typ	Max	Unit	Note
LED current	IL	-	30	40	mA	(2)
LED voltage	VL	-	16.8	19.2	V	
Operating LED life time	Hr	-	20000	15000	Hour	(1)(2)

Note (1) LED life time (Hr) can be defined as the time in which it continues to operate under the condition: Ta=25±3 ℃, typical IL value indicated in the above table until the brightness becomes less than 50%.

Note (2) The “LED life time” is defined as the module brightness decrease to 50% original brightness at Ta=25℃and IL=20mA.

5.0 OUTINE DIMENSION

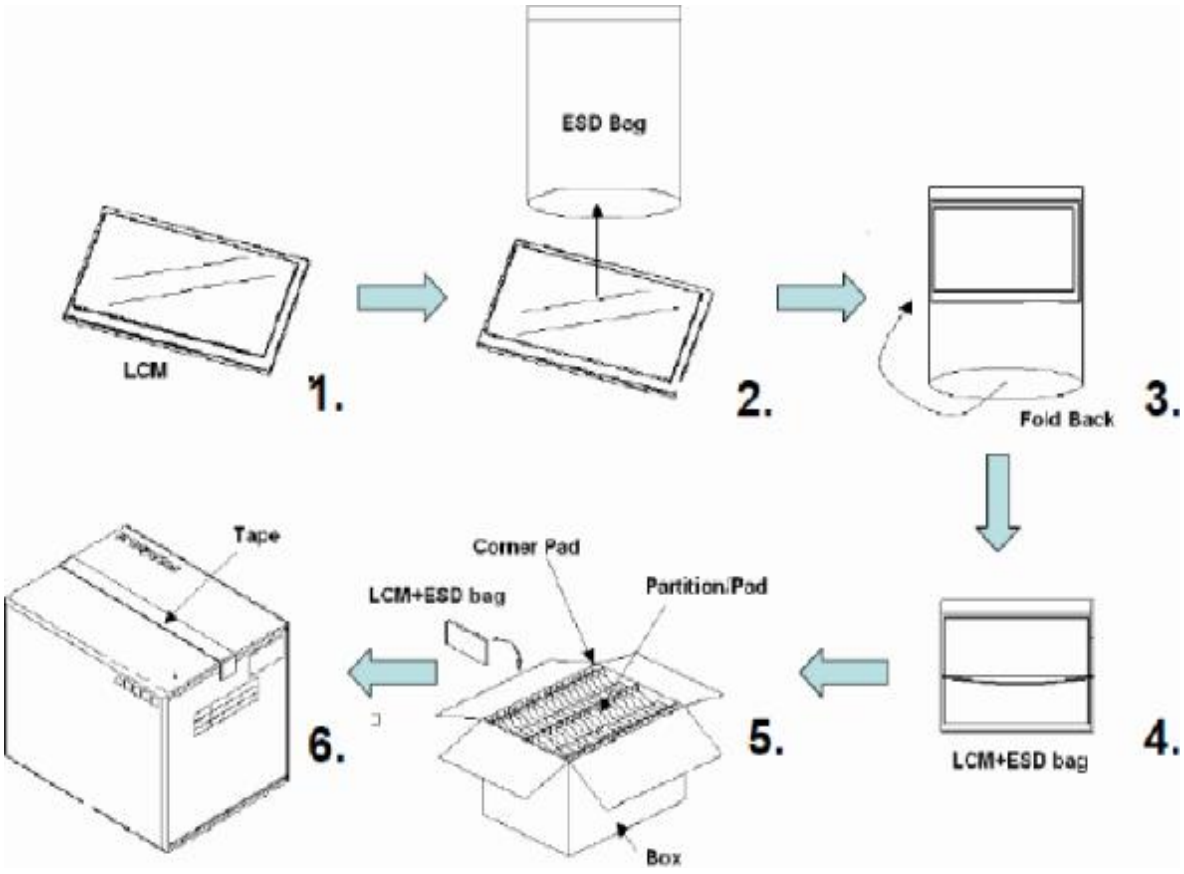


6.0 PACKAGE SPECIFICATION

6.1 Packing form

LCM Model	LCM Qty. in the box	Inner Box Size (mm)	Notice
	TDB	TDB	

6.2 Packing assembly drawings11.2 Packing assembly drawings



Items	Material	Notice
Box	Corrugated Paper Board	AB Flute
Partition/Pad	Corrugated Paper Board	A/B Flute
Corner Pad	Corrugated Paper Board	AB Flute
ESD bag	PE	

7.0 Items and Criteria：

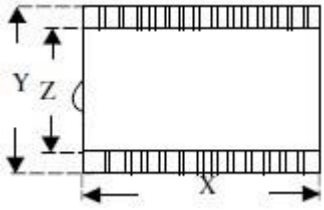
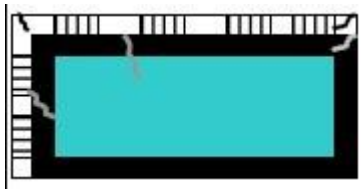
7.1 Guarantee

APEX warrants the quality of our products for ***1 year*** (from the date of delivery). If there are functional defects found during the period of warranty, the defective products would be replaced on a one-to-one ba Apex would not be responsible for any direct /indirect liabilities consequential to any parties.

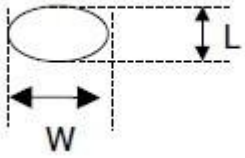
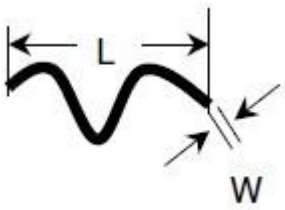
All the products should be stored or used as specified conditions described in these sheets. If module productions are not stored or used as specified conditions, herein, it will be void the ***1 year*** warranty(guarantee).

7.2 Visual inspection criterion in cosmetic

(1) Glass defect

Glass defect			
NO	Defect	Criteria	Remark
1	Dimension(Minor)	By engineering diagram	
2	Cracks(Major)	Extensive crack 【Reject】	

(2) LCM appearance defect

NO	Defect	Criteria		Remark
1	Round type(Minor)	Spec	Permissible Qty	1.ψ=(L+W)/2, L: Length, W: Width 2. Disregard if out of A.A. 
		ψ≦0.10mm	Disregard	
		0.10mm<ψ ≦ 0.20mm	**3**	
		0.20mm<ψ	**0**	
2	Line type(Minor)	Spec	Permissible Qty	1. L: Length, W: Width 2. Disregard if out of A.A. 
		W≦0.03mm	Disregard	
		L≦3.0mm and 0.03mm<W≦0.05mm	2	
		L≦3.0mm and 0.05mm<W≦0.10mm	1	
		W>0.10mm orL>3.0mm	0	
3	Polarizer dent(Minor)	Spec.	Permissible Qty	1.ψ=(L+W)/2 , L: Length, W: Width 2.Disregard if out of A.A.
		ψ≦0.20mm	Disregard	
		0.20mm<ψ≦ 0.30mm	2	
		0.30mm<ψ≦ 0.50mm	1	

(3) FPC

NO	Defect	Criteria	Remark
1	Copper peeling(Minor)	Copper peeling 【Reject】	
2	Golden finger	FPC golden finger broken, dead fold, indentation makes FPC surface broken 【Reject】 Tin plating layer(or gold plating) scratch, but not hurt circuit 【Accept】 Except circuit, other position scratch but not expose metal wire 【Accept】	
3	Pin	FPC PI layer delamination 【Reject】 Material and color are inconsistent with sample, FPC burrs 【Reject】 FPC Pin deformation but not affect function. 【Accept】 FPC Pin area is dirty 【Reject】 Other than FPC Pin area is dirty but not affect function 【Accept】	
4	Golden finger	Golden finger edge has burrs,foreign material【Reject】 Golden finger oxidation (dark), uneven electroplating, pinhole, foreign material 【Reject】 Golden finger soldering pad crack exceeds 1/3 length of soldering pad, and soldering pad crack exceed 2 Pins【Reject】 Golden finger tin plating(or gold plating)scratch, but not hurt circuit 【Accept】 Other than golden finger area scratch but not expose metal circuit 【Accept】	
5	FPC Silk printing	Ghosting, incomplete silk printing, wrong printing 【Reject】	
6	FPC Circuit line width	Line width deviation exceed 1/3 line width【Reject】	

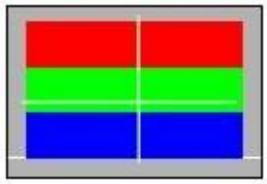
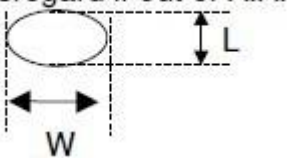
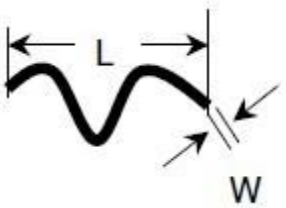
(4) Black tape

NO	Defect	Criteria	Remark
1	Shift(Minor)	IC exposed 【Reject】	
2	No black tape(Minor)	No black tape【Reject】	

(5) Silicon

NO	Defect	Criteria	Remark
1	Amount of silicon (Minor)	ITO exposed 【Reject】	

7.3 Visual inspection criterion in electrical display

<table>
<tr><th>NO</th><th>Defect</th><th colspan="3">Criteria</th><th>Remark</th></tr>
<tr><td>1</td><td>No display (Major)</td><td colspan="3">Not allowed</td><td>  </td></tr>
<tr><td>2</td><td>Missing line (Major)</td><td colspan="3">Not allowed</td><td>  </td></tr>
<tr><td>3</td><td>Darker or lighter Line (Major)</td><td colspan="3">Not allowed</td><td>  </td></tr>
<tr><td>4</td><td>Weak line(Major)</td><td colspan="3">By limited sample</td><td></td></tr>
<tr><td rowspan="3">5</td><td rowspan="3">Bright / Dark point (Minor)</td><td>Spec.</td><td colspan="2">Permissible Qty</td><td rowspan="3">1:1sub-pixel: 1R or 1G or1B
2:Point defect area≦1/2 sub pixel.</td></tr>
<tr><td>Bright point</td><td colspan="2">1</td></tr>
<tr><td>Dark point</td><td colspan="2">2</td></tr>
<tr><td rowspan="4">6</td><td rowspan="4">Round type (Minor)</td><td colspan="2">Spec</td><td>Permissible Qty</td><td rowspan="4">1.ψ=(L+W)/2, L: Length, W: Width
2. Disregard if out of A.A.
  </td></tr>
<tr><td colspan="2">ψ≦0.10mm</td><td>Disregard</td></tr>
<tr><td colspan="2">0.10mm<ψ≦ 0.20mm</td><td>3</td></tr>
<tr><td colspan="2">0.20mm<ψ</td><td>0</td></tr>
<tr><td rowspan="5">7</td><td rowspan="5">Line type (Minor)</td><td colspan="2">Spec.</td><td>Permissible Qty</td><td rowspan="5">1. L: Length, W: Width
2. Disregard if out of A.A.
  </td></tr>
<tr><td colspan="2">W≦0.03mm</td><td>Disregard</td></tr>
<tr><td colspan="2">L≦3.0mm and 0.03mm<W≦0.05mm</td><td>2</td></tr>
<tr><td colspan="2">L≦3.0mm and 0.05mm<W≦0.10mm</td><td>1</td></tr>
<tr><td colspan="2">W>0.10mm or L>3.0mm</td><td>0</td></tr>
<tr><td>8</td><td>Mura (Minor)</td><td colspan="3">By 5% ND filter invisible</td><td></td></tr>
</table>